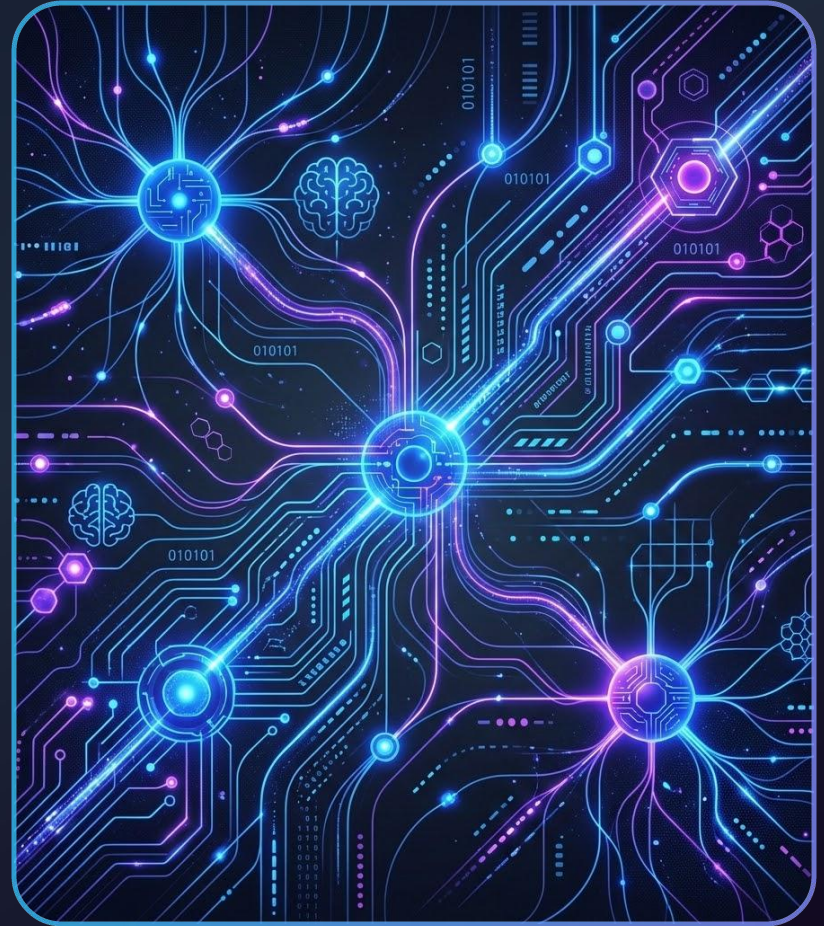


WEEK 08

How **Encoding** Supports Hybrid Model

Patrina Pun

Undergraduate



What is Angle Encoding & a Hybrid Model?

Angle Encoding

A core technique in quantum computing used to translate classical data into quantum states.

How it works:

Maps classical numerical values directly onto the rotation angles of quantum gates, preparing them as operable quantum states.

Hybrid Model

A computational framework where classical and quantum computers work together to solve complex problems.

Division of Labor:

Classical: Handles data formatting, control logic, and optimization.

Quantum: Manages specific calculations leveraging superposition and entanglement.

How does the Hybrid Model Work?

01

Classical Setup

The classical computer prepares the data or defines the problem parameters and sends them to the quantum processor.

02

Quantum Proc.

The quantum processor runs a circuit using those parameters. The quantum circuit applies parameterized quantum gates to the encoded qubits. Superposition and entanglement allows the circuit to learn complex feature relationships.

03

Measurement

The quantum state is measured, collapsing it into classical output (numbers or bits).

04

Evaluation & Opt.

The classical computer takes that output, evaluates its quality. The classical optimizer computes the loss, updates the trainable parameters using backpropagation or gradient-based optimization, and prepares the next quantum circuit.

05

Repeat Cycle

The updated parameters go back to the quantum processor and the cycle continues until the result converges on a good solution.

How does Encoding support a Hybrid Model?

01

Classical-Quantum Bridge

Angle encoding supports hybrid models by providing the link between classical and quantum computation. Classical clinical features are first normalized and converted into qubit rotation angles, allowing the quantum circuit to process information that originates from a classical computer. After quantum processing, the measured outputs are returned to the classical model, enabling both systems to work together during prediction and training.

02

NISQ

Angle encoding supports hybrid models by efficiently embedding classical data into quantum circuits using simple qubit rotation gates. Because relatively few quantum operations are required, the overall circuit remains shallow, reducing the effects of noise on today's Noisy Intermediate-Scale Quantum (NISQ) devices. This makes angle encoding a practical choice for integrating quantum layers into hybrid machine learning models.